Financial Mathematics 32000

Lecture 1

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2026 March 23

FINM 33000 and FINM 32000

This is the second course in the sequence.

- Fall 2025: Options
- Spring 2026: Numerical methods for options
- "Options" is meant in a broad sense: the pricing and hedging of options and other financial *derivative contracts*
- A *derivative security* or *derivative contract* is a financial instrument whose payoff is defined in terms of an *underlying* (e.g.: A security such as a stock/bond. An index. An interest rate.)

Why do we need numerical methods

Because we do not have simple exact formulas

- when the *dynamics* of the underlying risks are too complicated. For example, when working outside the class of Gaussian models (such as BM and GBM), simple exact formulas are less common. Why isn't GBM enough? Because it's not consistent with the **volatility skew**.
- when the *contract* to be priced/hedged is too complicated. For example, path-dependent options are typically harder to price than European-style options. Path-dependent options include **American**-style options, which can be exercised before expiry.

The implied volatility skew

American options

UNIT 0: Analytic approximation

Realized Volatility

Realized variance of S, sampled at interval Δt, from time 0 to time T can be defined, using log-returns by letting $t_n = n\Delta t$ and $T = t_N$ and

$$RVar = \frac{1}{T}\sum_{n=0}^{N-1}\left(\log\frac{S_{t_{n+1}}}{S_{t_n}}\right)^2$$

Alternatively could use simple returns, letting $\Delta S = S_{t_{n+1}} - S_{t_n}$ and

$$RVar = \frac{1}{T}\sum_{n=0}^{N-1}\left(\frac{\Delta S}{S_{t_n}}\right)^2$$

Alternatively, subtract the sample mean. **Realized volatility** of S is

$$RVol = \sqrt{RVar}$$

If $\mathrm{d}S_t = \mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$, then $RVol \to \sigma$ as $\Delta t \to 0$.

Black Scholes

Define the function C^{BS} for $X > 0, K > 0, \sigma > 0, t < T$ by

$$C^{BS}(X, t, K, T, R_{grow}, R_{disc}, \sigma) := e^{-R_{disc}(T-t)}\big[FN(d_1) - KN(d_2)\big],$$

and

$$F := Xe^{R_{grow}(T-t)}$$

and

$$d_{1,2} := d_{+,-} := \frac{\log(F/K)}{\sigma\sqrt{T-t}} \pm \frac{\sigma\sqrt{T-t}}{2}.$$

Recall: if $\mathrm{d}X_t = R_{grow}X_t\mathrm{d}t + \sigma X_t\mathrm{d}W_t$ where W is $\mathbb{P}$-BM. then

$$e^{-R_{disc}(T-t)}\mathbb{E}_t(X_T - K)^+ = C^{BS}(X_t, t, K, T, R_{grow}, R_{disc}, \sigma)$$

so C^{BS} gives the time-t price of a T-expiry K-strike call on S.

Implied Volatility

Given a time-t price C_t of a European call option (K, T) on a no-div stock S, the time-t **implied volatility** is the σ such that

$$C_t = C^{BS}(S_t, t, K, T, r, r, \sigma)$$

where C^{BS} is the Black-Scholes formula and r is the interest rate.

- This exists and is unique, if $(S_t - Ke^{-r(T-t)})^+ < C_t < S_t$
- The bigger the dollar price C_t, the bigger the implied vol σ_{imp}

  $\sigma \to 0 \quad C_t \quad \sigma \to \infty$
- Gives us another way quoting an option price. Instead of saying the option is trading at \$`x.xx`, can say it's trading at `yy`% vol.

Implied Volatility

For **general** underlying Y_T, let F_t be time-t forward price. Recall:

- Contract with payoff $Y_T - F_t$ has time-t price 0. Thus $F_t = \mathbb{E}_t Y_T$
- $F_t - K = (C_t - P_t)e^{r(T-t)}$ where P_t is time-t price of (K, T) European put. Thus $F_t = K_*$ if K_* is the strike where $C_t = P_t$.

Implied volatility of a European (K, T) call on X is the σ such that

$$C_t = C^{BS}(F_t, t, K, T, 0, r, \sigma)$$

Implied volatility of a European (K, T) **put** on X is the σ such that

$$P_t = P^{BS}(F_t, t, K, T, 0, r, \sigma)$$

where $P^{BS} = e^{-R_{disc}(T-t)}\left[K N(-d_2) - F N(-d_1)\right]$ is B-S put formula.

- In frictionless markets, call implied vol = put implied vol

Interpretations

Interpretations of time-t implied volatility

- Intuitively, implied vol is in some sense the market's forward-looking expectation of "realized volatility" from time t until T, along paths that go near K.
- A language / a metric / a scale in which to quote option prices. Instead of quoting in dollars, can quote in vol points. (Use of this language does not mean we actually believe the Black-Scholes assumptions!) Analogy: quoting a bond price as a yield-to-maturity does not mean we actually believe that interest rates are constant.

Volatility smile/skew

If the underlying truly has GBM dynamics with volatility σ then

$$C(K,T) = C^{BS}(K,T,\sigma) \qquad \text{for all } (K,T)$$

Hence

$$\sigma_{imp}(K,T) = \sigma \qquad \text{for all } (K,T)$$

However, empirically,

- Plotting σ_{imp} against K, you typically see not a flat line, but rather a volatility *smile* or a volatility *skew*.
- Also, σ_{imp} varies wrt T; implied volatility has a *term structure*.
- The function $\sigma_{imp}(K,T)$ is called the implied vol *surface* or *skew*.

Volatility surface: SPX, by strike

SPX Index | 90) Asset | 91) Actions | 92) Views | 93) Settings | Volatility Surface

S&P 500 INDEX 4720.29 USD Bloomberg Mid As of 12-Jan-2022 15:35

1) Vol Table | 2) 3D Surface | 3) Term | 4) Skew | 5) Dividends | 6) Prices

Strike Listed 10) Edit ☑ Fwd ☑ Strikes

Exp Date	ImpFwd	4670.0	4675.0	4680.0	4685.0	4690.0	4695.0	4700.0	4705.0	4710.0	4715.0	4720.0	4725.0	4730.0	4735.0	4740.0	4745.0	4750.0	4755.0	4760.0	4765.0	4770.0	4775.0
		4670.0	4675.0	4680.0	4685.0	4690.0	4695.0	4700.0	4705.0	4710.0	4715.0	4720.0	4725.0	4730.0	4735.0	4740.0	4745.0	4750.0	4755.0	4760.0	4765.0	4770.0	4775.0
14 Jan 2022	4719.93	18.92	18.66	18.40	18.14	17.88	17.61	17.35	17.07	16.79	16.49	16.18	15.89	15.67	15.53	15.41	15.29	15.16	15.04	14.92	14.82	14.73	14.66
18 Jan 2022	4719.83	13.84	13.68	13.52	13.36	13.19	13.01	12.84	12.65	12.47	12.27	12.07	11.88	11.70	11.57	11.46	11.35	11.24	11.14	11.04	10.95	10.87	10.80
19 Jan 2022	4719.86	14.40	14.24	14.07	13.90	13.72	13.54	13.36	13.17	12.98	12.78	12.58	12.37	12.18	12.01	11.87	11.74	11.62	11.50	11.38	11.27	11.17	11.08
21 Jan 2022	4719.69	14.83	14.64	14.46	14.27	14.08	13.89	13.70	13.51	13.32	13.14	12.95	12.77	12.60	12.43	12.27	12.12	11.97	11.83	11.70	11.58	11.47	11.37
24 Jan 2022	4719.61	14.62	14.45	14.29	14.12	13.95	13.78	13.61	13.44	13.28	13.11	12.95	12.78	12.62	12.47	12.31	12.16	12.02	11.88	11.75	11.62	11.50	11.38
26 Jan 2022	4719.58	15.66	15.49	15.33	15.16	14.99	14.83	14.66	14.49	14.33	14.16	13.99	13.83	13.67	13.51	13.35	13.20	13.04	12.90	12.75	12.61	12.48	12.35
28 Jan 2022	4719.11	16.16	16.00	15.84	15.67	15.51	15.34	15.17	15.01	14.84	14.67	14.51	14.34	14.18	14.01	13.85	13.70	13.54	13.39	13.24	13.09	12.95	12.81
31 Jan 2022	4718.92	15.55	15.40	15.25	15.09	14.94	14.78	14.63	14.47	14.32	14.16	14.00	13.85	13.70	13.54	13.39	13.24	13.10	12.95	12.81	12.67	12.54	12.41
2 Feb 2022	4718.98	15.76	15.61	15.46	15.31	15.15	15.00	14.85	14.70	14.54	14.39	14.24	14.09	13.93	13.79	13.64	13.49	13.35	13.20	13.06	12.93	12.79	12.66
4 Feb 2022	4718.25	15.92	15.78	15.63	15.48	15.33	15.18	15.04	14.89	14.74	14.59	14.44	14.29	14.15	14.00	13.86	13.72	13.57	13.44	13.30	13.16	13.03	12.90
7 Feb 2022	4718.32	15.46	15.32	15.18	15.04	14.90	14.76	14.62	14.48	14.34	14.20	14.06	13.92	13.78	13.64	13.50	13.37	13.23	13.10	12.97	12.84	12.72	12.59
9 Feb 2022	4717.54	15.59	15.45	15.32	15.18	15.04	14.90	14.76	14.62	14.48	14.34	14.20	14.07	13.93	13.79	13.66	13.52	13.39	13.26	13.13	13.01	12.88	12.76
11 Feb 2022	4717.17	15.79	15.66	15.52	15.39	15.25	15.12	14.98	14.85	14.71	14.57	14.44	14.31	14.17	14.04	13.91	13.78	13.65	13.52	13.39	13.27	13.14	13.02
18 Feb 2022	4715.36	15.70	15.58	15.45	15.33	15.20	15.07	14.95	14.82	14.69	14.57	14.44	14.32	14.19	14.07	13.95	13.82	13.70	13.58	13.46	13.35	13.23	13.12
25 Feb 2022	4714.54	15.80	15.68	15.57	15.45	15.33	15.22	15.10	14.99	14.87	14.75	14.64	14.52	14.41	14.29	14.18	14.07	13.95	13.84	13.73	13.62	13.52	13.41
28 Feb 2022	4714.09	15.63	15.51	15.40	15.29	15.17	15.06	14.95	14.83	14.72	14.61	14.50	14.38	14.27	14.16	14.05	13.94	13.83	13.72	13.61	13.51	13.40	13.30
18 Mar 2022	4711.30	16.36	16.26	16.16	16.06	15.96	15.86	15.76	15.66	15.56	15.46	15.36	15.27	15.17	15.07	14.97	14.87	14.77	14.68	14.58	14.49	14.39	14.30
31 Mar 2022	4710.35	16.73	16.63	16.54	16.45	16.36	16.26	16.17	16.08	15.98	15.89	15.80	15.71	15.61	15.52	15.43	15.34	15.24	15.15	15.06	14.97	14.88	14.79
14 Apr 2022	4709.51	17.03	16.94	16.85	16.77	16.68	16.59	16.50	16.42	16.33	16.24	16.15	16.07	15.98	15.89	15.81	15.72	15.63	15.55	15.46	15.38	15.30	15.21
29 Apr 2022	4709.62	17.39	17.31	17.23	17.15	17.07	16.99	16.91	16.83	16.75	16.67	16.59	16.51	16.43	16.35	16.27	16.19	16.11	16.03	15.95	15.87	15.79	15.72
20 May 2022	4706.54	17.83	17.75	17.68	17.61	17.53	17.46	17.39	17.31	17.24	17.16	17.09	17.02	16.94	16.87	16.80	16.72	16.65	16.58	16.50	16.43	16.36	16.29
31 May 2022	4705.83	17.94	17.86	17.79	17.72	17.65	17.58	17.51	17.44	17.37	17.29	17.22	17.15	17.08	17.01	16.94	16.87	16.80	16.73	16.66	16.59	16.52	16.45
17 Jun 2022	4703.73	18.31	18.24	18.17	18.11	18.04	17.97	17.90	17.83	17.77	17.70	17.63	17.56	17.50	17.43	17.36	17.30	17.23	17.16	17.10	17.03	16.96	16.90
30 Jun 2022	4703.82	18.46	18.40	18.33	18.27	18.20	18.14	18.07	18.00	17.94	17.87	17.81	17.74	17.68	17.61	17.55	17.48	17.42	17.35	17.28	17.22	17.16	17.09
15 Jul 2022	4703.38	18.62	18.56	18.49	18.43	18.36	18.30	18.23	18.17	18.10	18.04	17.97	17.91	17.84	17.78	17.72	17.65	17.59	17.53	17.46	17.40	17.34	17.28
19 Aug 2022	4701.68	19.08	19.02	18.96	18.90	18.84	18.78	18.72	18.66	18.60	18.54	18.48	18.42	18.36	18.30	18.24	18.18	18.12	18.06	18.00	17.94	17.89	17.83
16 Sep 2022	4699.69	19.34	19.28	19.23	19.17	19.11	19.05	19.00	18.94	18.88	18.83	18.77	18.71	18.66	18.60	18.54	18.49	18.43	18.37	18.32	18.26	18.21	18.15
30 Sep 2022	4700.41	19.52	19.47	19.41	19.36	19.30	19.25	19.19	19.14	19.08	19.03	18.98	18.92	18.87	18.81	18.76	18.70	18.65	18.59	18.54	18.49	18.43	18.38
21 Oct 2022	4700.74	19.67	19.62	19.57	19.51	19.46	19.40	19.35	19.30	19.24	19.19	19.14	19.08	19.03	18.97	18.92	18.87	18.81	18.76	18.71	18.66	18.60	18.55

Option Pricing (OVME) Legend Zoom 100%

99) Quick Pricer

SN 417340 EST GMT-5:00 G721-6557-166 12-Jan-2022 15:56:46

Volatility surface: SPX, by moneyness

SPX Index | Asset | Actions | Views | Settings | Volatility Surface

S&P 500 INDEX 4719.19 USD Bloomberg Mid As of 12-Jan-2022 15:40

Vol Table | 3D Surface | Term | Skew | Dividends | Prices

Moneyness | Listed | Edit | Fwd | Strikes

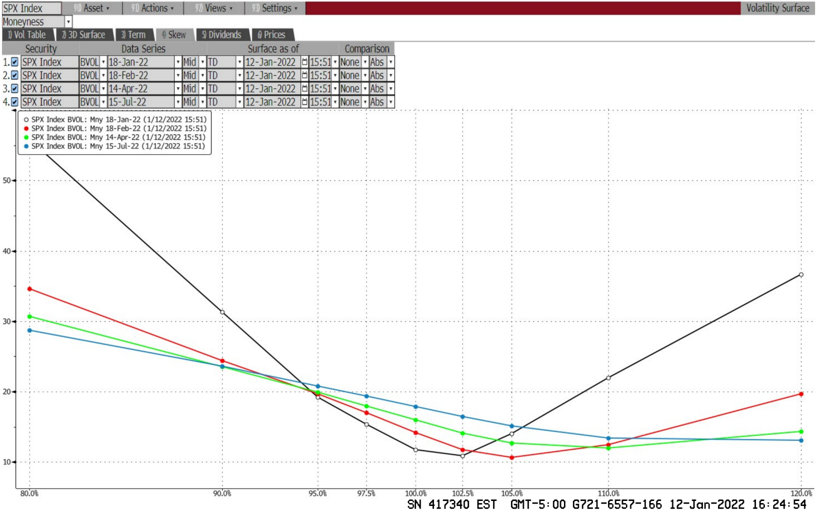
Exp Date	ImpFwd	80.0%	90.0%	95.0%	97.5%	100.0%	102.5%	105.0%	110.0%	120.0%
		3775.4	4247.3	4483.2	4601.2	4719.2	4837.2	4955.1	5191.1	5663.0
14 Jan 2022	4718.94	76.98	45.47	27.95	21.98	16.30	15.63	22.34	32.87	51.93
18 Jan 2022	4718.84	56.42	31.81	19.91	15.90	12.16	11.10	14.47	22.08	36.64
19 Jan 2022	4718.80	54.06	30.48	20.04	16.48	12.65	11.10	14.13	21.08	34.82
21 Jan 2022	4718.59	50.26	29.54	20.77	17.01	13.02	11.03	13.49	20.00	32.02
24 Jan 2022	4718.52	46.25	27.57	20.05	16.67	13.00	10.69	12.28	18.03	29.06
26 Jan 2022	4718.49	44.31	27.95	21.01	17.73	14.04	11.31	12.04	17.31	27.60
28 Jan 2022	4718.04	42.99	28.07	21.47	18.21	14.55	11.60	11.86	16.63	26.43
31 Jan 2022	4717.85	40.80	26.61	20.51	17.47	14.04	11.20	11.19	15.39	24.87
2 Feb 2022	4717.91	39.91	26.55	20.65	17.65	14.27	11.40	11.12	15.07	24.06
4 Feb 2022	4717.17	39.32	26.44	20.75	17.79	14.47	11.62	11.07	14.50	23.30
7 Feb 2022	4717.25	37.68	25.47	20.09	17.25	14.08	11.35	10.71	13.78	22.31
9 Feb 2022	4716.46	37.05	25.38	20.17	17.37	14.23	11.51	10.74	13.53	21.76
11 Feb 2022	4716.10	36.55	25.33	20.29	17.53	14.46	11.77	10.81	13.27	21.23
18 Feb 2022	4714.26	34.84	24.65	19.93	17.33	14.45	11.91	10.80	12.53	19.73
25 Feb 2022	4713.46	33.92	24.32	19.82	17.33	14.66	12.26	11.03	12.15	18.56
28 Feb 2022	4713.02	33.22	23.91	19.51	17.11	14.51	12.15	10.93	11.88	18.12
18 Mar 2022	4710.20	32.16	23.89	19.84	17.67	15.37	13.23	11.92	11.94	16.13
31 Mar 2022	4709.28	31.35	23.79	19.99	17.95	15.81	13.75	12.34	11.96	15.15
14 Apr 2022	4708.41	30.83	23.70	20.09	18.18	16.17	14.26	12.83	12.09	14.34
29 Apr 2022	4708.51	30.33	23.69	20.28	18.47	16.59	14.78	13.34	12.27	13.77
20 May 2022	4705.41	29.80	23.69	20.50	18.81	17.09	15.41	14.01	12.65	13.34
31 May 2022	4704.69	29.46	23.59	20.51	18.89	17.22	15.61	14.23	12.77	13.26
17 Jun 2022	4702.57	29.27	23.68	20.75	19.21	17.63	16.09	14.74	13.12	13.37
30 Jun 2022	4702.67	29.05	23.68	20.84	19.34	17.80	16.29	14.95	13.26	13.26
15 Jul 2022	4702.22	28.83	23.74	21.01	19.51	17.98	16.52	15.25	13.52	13.05
19 Aug 2022	4700.52	28.57	23.86	21.30	19.90	18.48	17.12	15.89	14.10	13.22
16 Sep 2022	4698.54	28.32	23.87	21.43	20.11	18.77	17.47	16.28	14.47	13.32
30 Sep 2022	4699.26	28.27	23.93	21.55	20.26	18.94	17.66	16.48	14.64	13.42
21 Oct 2022	4699.58	28.18	23.89	21.58	20.38	19.17	17.94	16.76	14.84	13.60

Option Pricing (OVME) | Legend | Zoom 100%

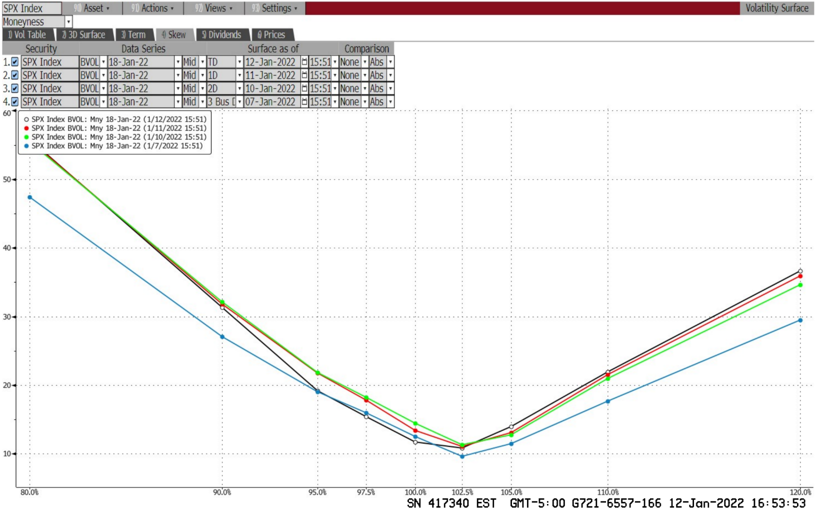
Quick Pricer

SN 417340 EST GMT-5:00 G721-6557-166 12-Jan-2022 15:58:14

Volatility skews: SPX, by moneyness, 4 expirations



Volatility skews: SPX, by moneyness, 4 observations



Volatility smile/skew: EBAY

EBAY US $ * ↓ 74.61 -1.59 B 1s Equity OCM

DELAY 12:26 Vol 5,752,860 Op 76.6 Q Hi 76.68 Q Lo 74.07 Q ValTrd 433.120m

OPTION MONITOR 1 COMP Center: 74.12 1 <GO> to Edit Spreadsheet

BID d	ASK d	IVBD	IVAS	IVMD	OPIN d		BID d	ASK d	IVBD	IVAS	IVMD	OPIN d
Bid Price	Ask Price	Imp Volat Bid	Imp Volat Ask	Imp Volat Mid	Open Intrst	EBAY <-CALLS PUTS->	Bid Price	Ask Price	Imp Volat Bid	Imp Volat Ask	Imp Volat Mid	Open Intrst
74.610	74.620					QXB JUL4	74.610	74.620				
44.60	44.80	N.A.	69.07	N.A.	335	1) 30 16)		.05	N.A.	74.86	69.02	478
39.60	39.80	N.A.	70.07	54.16	305	2) 35 17)		.05	N.A.	62.87	57.97	453
34.70	34.80	N.A.	57.73	36.43	378	3) 40 18)		.05	N.A.	52.53	48.44	1,169
29.70	29.90	N.A.	52.91	46.90	355	4) 45 19)	.05	.10	43.44	47.80	45.89	735
24.80	25.00	37.23	46.40	42.65	514	5) 50 20)	.15	.20	41.55	43.68	42.61	1,559
20.00	20.20	36.56	41.46	39.18	889	6) 55 21)	.30	.35	38.88	40.05	39.50	2,205
						XBA JUL4						
15.40	15.50	34.85	37.76	36.34	1,931	7) 60 22)	.65	.75	35.56	37.06	36.31	4,757
11.10	11.30	33.40	34.36	33.88	3,851	8) 65 23)	1.40	1.45	33.46	34.45	33.95	9,285
7.40	7.60	31.33	32.79	32.06	3,313	9) 70 24)	2.65	2.75	31.84	32.56	32.20	4,905
4.50	4.60	29.70	31.03	30.37	7,000	10) 75 25)	4.70	4.80	30.15	30.81	30.48	2,415
2.45	2.55	29.16	29.88	29.52	2,166	11) 80 26)	7.60	7.80	28.98	30.39	29.68	619
1.20	1.30	28.78	29.25	29.02	1,429	12) 85 27)	11.40	11.60	27.94	29.89	28.92	209
.55	.65	28.82	29.50	29.16	1,115	13) 90 28)	15.70	15.90	27.27	30.24	28.80	
.20	.30	27.69	30.02	28.92	16	14) 95 29)	20.40	20.70	N.A.	32.77	29.60	
.10	.15	28.66	30.60	29.69	305	15) 100 30)	25.30	25.50	N.A.	32.83	30.88	

Volatility smile/skew: GOOG

GOOG US $ ↑555.40 +.50 Q555.20/555.37Q 1x1

At 12:19 d Vol 1,878,854 O 559.62Q H 562.00P L 552.95P Val 1.048B

GOOG US Equity | 95) Templates | 96) Actions | 97) Expiry | Option Monitor: Option Monitor

GOOGLE INC-C ↑555.4001 .5001 .0901% 555.20 / 555.3701 Hi 562.00 Lo 552.95 Volm 1878854 HV .00

Calc Mode Center 555.00 Strikes 19 Exch US Composit 92) Next Earnings(EM) 04/16/14 C

81) Center Strike 82) Calls/Puts 83) Calls 84) Puts 85) Term Structure

	Calls								Strike		Puts							
	Ticker	Bid	Ask	Last	IVM	DM	Volm	OInt			Ticker	Bid	Ask	Last	IVM	DM	Volm	OInt
	19 Apr 14 (10d); CSize 100; IDiv .77; R .12; IFwd 556.55								19		19 Apr 14 (10d); CSize 100; IDiv .77; R .12; IFwd 556.55							
1)	GOOG 4 C510	48.20	50.80	50.00	58.97	.84	2	34	510.00	58)	GOOG 4 P510	3.20	3.60	3.08	55.30	-.14	69	203
2)	GOOG 4 C515	43.70	46.80	43.00y	57.94	.82		37	515.00	59)	GOOG 4 P515	4.10	4.40	4.24	55.20	-.17	28	304
3)	GOOG 4 C520	39.90	40.60	44.40	53.44	.81	11	35	520.00	60)	GOOG 4 P520	4.90	5.30	4.90	54.18	-.20	13	485
4)	GOOG 4 C525	36.20	37.70	40.00	55.38	.76	1	58	525.00	61)	GOOG 4 P525	6.00	6.40	6.14	53.70	-.23	27	163
5)	GOOG 4 C530	32.40	33.50	34.00	54.21	.73	1	76	530.00	62)	GOOG 4 P530	7.10	7.70	7.05	52.96	-.27	14	183
6)	GOOG 4 C535	28.90	29.80	30.56	53.31	.69	45	73	535.00	63)	GOOG 4 P535	8.70	9.10	7.80	52.84	-.30	45	348
7)	GOOG 4 C540	25.60	26.50	27.52	53.30	.65	12	214	540.00	64)	GOOG 4 P540	10.40	10.80	9.91	52.60	-.35	26	249
8)	GOOG 4 C545	22.50	23.30	23.70	52.81	.61	18	168	545.00	65)	GOOG 4 P545	12.10	12.70	12.30	51.82	-.39	17	148
9)	GOOG 4 C550	19.60	20.20	20.00	52.13	.56	168	358	550.00	66)	GOOG 4 P550	14.30	14.80	13.80	51.63	-.44	84	155
10)	GOOG 4 C555	17.00	17.50	17.95	51.84	.52	192	256	555.00	67)	GOOG 4 P555	16.60	17.10	16.50	51.18	-.48	41	159
11)	GOOG 4 C560	14.60	15.00	15.41	51.29	.47	472	694	560.00	68)	GOOG 4 P560	19.20	19.70	18.61	51.09	-.53	101	113
12)	GOOG 4 C565	12.40	13.00	13.14	51.28	.43	115	321	565.00	69)	GOOG 4 P565	21.80	22.50	20.50	50.64	-.58	1	29
13)	GOOG 4 C570	10.50	10.70	10.51	50.74	.38	154	652	570.00	70)	GOOG 4 P570	24.90	25.70	23.82	50.26	-.62	1	51
14)	GOOG 4 C575	8.80	9.20	9.20	50.26	.34	244	317	575.00	71)	GOOG 4 P575	28.00	28.80	31.60y	49.87	-.67		8
15)	GOOG 4 C580	7.30	7.80	7.50	50.28	.30	124	431	580.00	72)	GOOG 4 P580	31.40	32.30	29.10	49.19	-.71	11	47
16)	GOOG 4 C585	5.90	6.30	6.43	50.02	.25	28	211	585.00	73)	GOOG 4 P585	34.80	36.00	38.00y	48.78	-.75		11
17)	GOOG 4 C590	4.80	5.30	5.07	50.20	.22	71	333	590.00	74)	GOOG 4 P590	37.70	40.20	36.23	46.97	-.80	8	21
18)	GOOG 4 C595	3.90	4.40	4.15	50.05	.19	91	136	595.00	75)	GOOG 4 P595	41.60	44.10	40.26	44.85	-.84	8	13
19)	GOOG 4 C600	3.20	3.60	3.39	49.94	.16	80	837	600.00	76)	GOOG 4 P600	45.90	48.30	63.95y	44.80	-.87		35
	17 May 14 (38d); CSize 100; IDiv .23; R .16; IFwd 556.62								19		17 May 14 (38d); CSize 100; IDiv .23; R .16; IFwd 556.62							
20)	GOOG 5 C510	52.10	54.80	51.30y	36.10	.79		15	510.00	77)	GOOG 5 P510	6.70	7.20	5.80	33.61	-.20	6	323

93) Default color legend Zoom 90%

Volatility surface: GOOG

4/9 4/16 4/19 5/17

GOOG US $ ↓556.80 +1.90 P556.65/556.84Q 2x1

At 12:32 d Vol 1,926,365 O 559.62Q H 562.00P L 552.95P Val 1.075B

GOOG US Equity 95) Templates 96) Actions 97) Expiry Option Monitor: Option Monitor

GOOGLE INC-C ↓556.80 1.90 .3424% 556.6501 / 556.8401 Hi 562.00 Lo 552.95 Volm 1926365 HV .00

Calc Mode Center 555.00 Strikes 19 Exch US Composit 92) Next Earnings(EM) 04/16/14 C

81) Center Strike 82) Calls/Puts 83) Calls 84) Puts 85) Term Structure

Expiry	19 Apr 2014				17 May 2014				21 Jun 2014				20 Sep 2014				17 Jan 2015			
Calls/Puts	Calls		Puts		Calls		Puts		Calls		Puts		Calls		Puts		Calls		Puts	
Strike	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM
505.00	.91	51.46	-.11	56.24	.83	33.50	-.17	34.07	.80	28.63	-.20	28.79	.74	26.46	-.26	26.02	.71	25.49	-.29	25.08
510.00	.88	50.89	-.14	55.57	.81	33.13	-.20	34.25	.77	28.55	-.23	28.69	.72	26.38	-.28	26.03	.70	25.20	-.30	24.56
515.00	.85	51.33	-.16	55.34	.79	32.84	-.22	33.86	.75	28.26	-.25	28.51	.70	26.34	-.29	25.96	.68	25.16	-.32	24.68
520.00	.82	51.42	-.19	54.24	.76	33.23	-.24	33.63	.73	28.38	-.27	28.43	.68	26.45	-.31	25.86	.66	25.12	-.33	24.60
525.00	.79	51.53	-.22	54.26	.73	33.14	-.27	33.50	.70	28.74	-.30	28.37	.66	26.29	-.33	25.68	.65	25.21	-.35	24.43
530.00	.75	51.68	-.26	53.59	.70	33.56	-.30	33.37	.67	28.54	-.32	28.27	.64	26.18	-.35	25.61	.63	24.98	-.37	24.60
535.00	.71	52.68	-.29	53.02	.67	33.28	-.33	33.16	.65	28.54	-.35	28.08	.62	26.24	-.37	25.49	.62	25.14	-.38	24.31
540.00	.67	52.50	-.33	52.64	.63	33.20	-.37	32.79	.62	28.27	-.38	27.94	.60	26.08	-.39	25.46	.60	25.08	-.40	24.43
545.00	.62	52.27	-.38	52.45	.60	32.95	-.40	32.84	.59	28.20	-.41	27.78	.58	26.14	-.42	25.45	.58	25.15	-.42	24.24
550.00	.58	51.82	-.42	51.92	.57	32.76	-.43	32.58	.56	28.04	-.44	27.75	.56	25.98	-.44	25.37	.57	24.96	-.43	24.37
555.00	.53	51.92	-.47	51.67	.53	32.61	-.47	32.44	.53	27.99	-.46	27.64	.54	26.03	-.46	25.25	.55	25.07	-.45	24.23
560.00	.48	50.99	-.52	51.18	.50	32.35	-.50	32.40	.51	27.99	-.49	27.55	.52	25.97	-.48	25.30	.53	25.01	-.47	24.27
565.00	.44	50.88	-.56	50.90	.46	32.34	-.54	32.05	.48	27.74	-.52	27.41	.50	25.95	-.50	25.28	.52	25.00	-.48	24.20
570.00	.39	50.26	-.61	50.53	.43	32.02	-.57	32.02	.45	27.70	-.55	27.48	.48	25.86	-.52	25.22	.50	24.83	-.50	24.31
575.00	.35	49.95	-.65	50.23	.39	31.85	-.61	31.75	.42	27.58	-.58	27.31	.46	25.80	-.54	25.13	.49	24.90	-.52	24.28
580.00	.30	49.76	-.69	50.05	.36	31.95	-.64	31.76	.39	27.49	-.61	27.20	.44	25.71	-.56	25.17	.47	24.86	-.53	24.03
585.00	.26	49.38	-.73	50.06	.33	31.86	-.67	31.66	.36	27.40	-.64	27.10	.42	25.63	-.58	25.28	.45	24.83	-.55	24.02
590.00	.23	49.41	-.76	51.33	.30	31.71	-.70	31.47	.34	27.28	-.66	27.28	.40	25.76	-.60	25.23	.44	24.78	-.57	23.95
595.00	.19	49.62	-.79	51.70	.27	31.30	-.73	31.21	.31	27.35	-.69	27.23	.38	25.89	-.62	25.18	.42	24.39	-.58	23.86
600.00	.16	49.15	-.82	51.71	.24	31.34	-.76	31.59	.29	27.29	-.71	27.10	.37	25.86	-.64	25.20	.41	24.59	-.60	23.87
605.00	.14	49.64	-.85	51.82	.22	31.06	-.78	31.40	.27	27.28	-.74	26.76	.35	25.87	-.66	25.24	.39	24.74	-.61	23.91
610.00	.12	49.72	-.88	50.09	.19	31.02	-.80	31.89	.24	27.21	-.76	27.17	.33	25.94	-.68	25.11	.38	24.65	-.63	23.79

93) Default color legend Zoom 85%

4/9 4/19 4/25 5/17

Volatility surface: AMZN

AMZN US $ ↓326.48 -.59 P326.36/326.51P 2x3
At 13:08 d Vol 2,908,935 O 328.47Q H 328.49D L 322.50Q Val 947.826M

AMZN US Equity | 95) Templates | 96) Actions | 97) Expiry | Option Monitor: Option Monitor

AMAZON.COM INC ↓326.48 -.59 -.1804% 326.360 / 326.51 Hi 328.49 Lo 322.50 Volm 2908935 HV 24.76

Calc Mode Center 326.66 Strikes 5 Exch US Composit 92) Next Earnings(EM) 04/25/14 E

81) Center Strike 82) Calls/Puts 83) Calls 84) Puts 85) Term Structure

Expiry	19 Apr 2014				17 May 2014				21 Jun 2014				19 Jul 2014				18 Oct 2014			
Calls/Puts	Calls		Puts		Calls		Puts		Calls		Puts		Calls		Puts		Calls		Puts	
Strike	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM	DM	IVM
280.00	.98	55.66	-.02	53.40	.86	48.25	-.14	46.67	.83	40.35	-.16	38.61	.81	36.77	-.18	35.63	.77	34.48	-.22	33.08
285.00	.98	47.38	-.03	50.74	.83	48.31	-.16	45.71	.81	39.40	-.19	38.04	.79	36.61	-.20	35.10	.75	34.06	-.24	32.75
290.00	.94	53.36	-.05	48.05	.81	47.17	-.19	44.94	.78	39.05	-.21	37.48	.76	36.34	-.23	34.44	.73	33.43	-.27	32.50
295.00	.93	47.70	-.06	45.69	.78	45.17	-.22	44.18	.75	38.71	-.24	37.02	.74	35.31	-.26	34.08	.71	32.87	-.29	32.11
300.00	.89	47.88	-.09	43.23	.75	44.56	-.25	43.42	.72	38.21	-.27	36.51	.71	34.87	-.29	34.00	.68	32.81	-.31	31.80
305.00	.85	45.62	-.13	41.09	.71	43.74	-.29	42.74	.69	37.40	-.31	36.07	.68	34.48	-.32	33.54	.66	32.30	-.34	31.68
310.00	.82	39.98	-.18	39.65	.67	43.80	-.33	42.16	.65	37.05	-.34	35.56	.65	33.86	-.35	33.14	.63	32.05	-.37	31.33
315.00	.74	38.38	-.26	38.31	.63	42.99	-.37	41.56	.61	36.59	-.38	35.35	.62	33.54	-.38	32.92	.61	32.05	-.39	31.12
320.00	.65	37.84	-.35	37.21	.58	42.54	-.42	41.19	.58	36.16	-.42	34.67	.58	33.56	-.42	32.66	.58	31.64	-.42	30.93
325.00	.54	36.78	-.46	36.31	.54	41.87	-.46	40.53	.54	35.78	-.46	34.46	.54	33.18	-.46	32.43	.55	31.49	-.45	30.69
330.00	.43	35.57	-.57	34.99	.49	41.35	-.51	40.00	.50	35.45	-.50	33.89	.51	33.00	-.49	32.16	.53	31.53	-.47	30.65
335.00	.32	34.82	-.69	34.19	.44	40.81	-.56	39.45	.46	35.01	-.54	33.67	.47	32.67	-.53	31.98	.50	31.29	-.50	30.59
340.00	.22	34.59	-.78	33.80	.40	40.58	-.60	39.57	.42	34.77	-.58	33.26	.44	32.63	-.57	31.62	.47	31.21	-.53	30.29
345.00	.15	34.47	-.85	34.14	.35	40.18	-.65	38.82	.39	34.63	-.61	33.73	.40	32.40	-.60	31.50	.45	31.09	-.56	30.05
350.00	.09	34.92	-.91	33.55	.31	40.07	-.70	38.37	.35	34.56	-.65	33.51	.37	32.30	-.63	31.43	.42	30.80	-.58	30.03
355.00	.06	35.90	-.97	28.83	.27	39.71	-.73	38.86	.31	34.42	-.70	32.51	.34	32.17	-.67	31.27	.40	30.93	-.61	30.04
360.00	.04	37.29	-.99	29.45	.24	39.64	-.77	38.67	.28	34.01	-.72	33.19	.31	32.09	-.70	31.04	.37	30.75	-.63	29.83
365.00	.03	38.73			.20	39.53	-.80	38.60	.25	33.81	-.75	32.76	.28	31.82	-.73	30.85	.35	30.49	-.66	29.60
370.00	.02	40.89	-1.0		.17	39.55	-.84	37.03	.22	33.96	-.79	31.97	.25	31.75	-.76	30.72	.32	30.37	-.68	29.49

93) Default color legend Zoom 100%

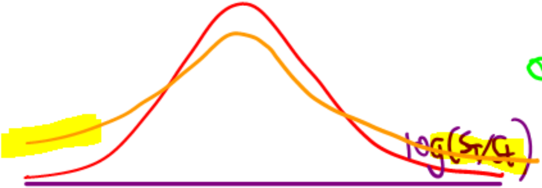
Volatility smile: possible explanations

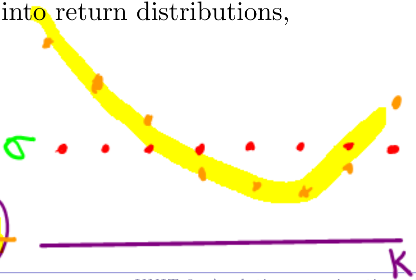
Why does the smile exist?

- The market prices options using a risk-neutral distribution of log-returns with *fatter tails* than the Normal (*excess kurtosis*).

What contributes to those fat tails?

- Clustering of volatility.
- Jumps introduce extreme outcomes into return distributions, especially short-term.





Volatility skew in equity markets

Why's the smile skewed? (Why fatter tails on *left*?) Physical causes:

- Negative correlation between price and instantaneous volatility. Empirically, instantaneous volatility increases as price decreases. This fattens the left tail of the distribution (negative skewness).
- Possibility of big down-jumps (crashes) also fattens the left tail.
- "The bull walks up the stairs, the bear jumps out the window."

Combined with risk preferences:

- "Supply and demand": Fear of downside leads to demand for protection/insurance, driving up the prices (hence implied vols) of OTM puts. And willingness to sell part of the upside leads (covered-) call writers to supply OTM calls, driving prices down.

Volatility skew: possible explanations

Skip

Why is the smile skewed? Another way to think about it:

- Implied volatility at strike K depends on the option price, which depends on expected future volatility – specifically, on expected future volatility *along paths near* K. Why? Future volatility along paths away from K does not help the option holder, because linear payoffs (e.g. zero, forward) are insensitive to vol. Only the *convex* part of the payoff brings benefits from volatility.
- Future volatility along paths near downside strikes is likely higher than future volatility along paths near upside strikes.
  So by the above logic, downside-strike options should trade higher above their intrinsic lower bound than upside-strike options do.

Volatility skew: implications

Presence of the volatility skew is inconsistent with GBM dynamics.

- Black-Scholes not adequate for pricing contracts such as forward-starting options, VIX options, etc, which are not just combinations of vanilla options. Consistent dynamic model needed.
- For simple purposes, using Black-Scholes with many different volatility inputs may be adequate (even if logically inconsistent):
  - Interpolating/extrapolating call or put prices at liquid strikes, to price a call or put at an illiquid strike
  - Detecting strikes at a given expiry that seem under/overpriced

  These purposes may just require fitting an implied vol curve.

The implied volatility skew

American options

UNIT 0: Analytic approximation

Pricing an American

- An *American* option on S with strike K and expiry T pays

$$\text{Put: } (K - S_\tau)^+ \qquad \text{Call: } (S_\tau - K)^+$$

  at the time of exercise τ, where $\tau \le T$ is a *stopping time* chosen by the holder. Not a stopping time: $\tau = \operatorname{argmin}_{t\in[0,T]} S_t$
- By definition the *stopping time* τ can be random, but for each t it must satisfy $\{\tau = t\} \in \mathcal{F}_t$; thus the decision on whether to exercise at time t must depend only on information that has been revealed up to and including time t.
- The ability to exercise early makes the contract path-dependent; its payout no longer depends only on the underlying at a particular time T.

Pricing an American: call options

In general, pricing early-exercisable contracts requires numerics.

An exception: if $r > 0$, then it is *never optimal to exercise early an American call on a non-dividend-paying stock.*

Model-independent reason: (with all expiries $= T$; all strikes $= K$; discount bond Z), the time-t (where $t < T$) values of

$$\text{American call} \geq \text{European call} \geq \text{Forward} = S_t - KZ_t > S_t - K$$

American is worth $>$ the exercise payoff $S_t - K$. So *don't exercise.*

- But what if you think S is going down? Shorting the stock beats exercising your call. By shorting you still get S_t, but you *delay* paying K, and if stock dips, you may pay *less* than K to cover.

Therefore, *American call price = European call price*

Pricing an American call: arbitrage argument

Arbitrage argument to show that American = European call price:

Let $r \geq 0$ and no dividends. If American > European then go short the American and long the European for an immediate profit.

- If the counterparty never exercises, then no further liabilities.
- If the counterparty exercises, then you are short a share and long a value of K dollars in the bank account, which becomes $\geq K$ by expiry. Covering the short at expiration requires $\leq K$ dollars, due to the European.

Since American > European leads to arbitrage, we must have *American call price = European call price.*

And *early exercise of the American cannot be strictly optimal.*

The implied volatility skew

American options

UNIT 0: Analytic approximation

Analytic approximation techniques

Analytic approximation techniques/heuristics help us to

- Analyze the accuracy of numerical methods
- Perform sanity checks on the output of numerical methods

Two of those analytic heuristics are:

- Taylor approximation
- Normal approximation

Taylor approximation

Postpone error analysis until we do finite difference methods.

A key tool in that error analysis will be Taylor approximation:

If f has $n+1$ continuous derivatives in a neighborhood of x_0 then

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + O(x - x_0)^{n+1}$$

as $x \to x_0$.

We may use Taylor approximation for two different purposes:

- To approximate prices and sensitivities – example on next page.
- To analyze the error in tree or finite difference calculation of prices and sensitivities

Interview question: price ATM option

No calculators allowed. Your interviewer says to you:

> *Spot is* 100. *No dividends. What's the price of a European-style* 1*-year at-the-money-forward (ATMF) vanilla option with* 20% *implied volatility?*

$= \mathbb{E}_0 S_T$

(ATMF at time 0 means $K = F_0$. Recall $F_0 = S_0 e^{rT}$ if no divs.)

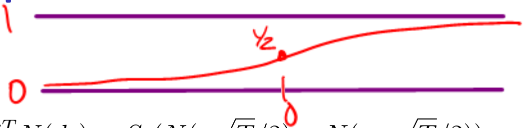
Answer: Black-Scholes call price is

$$S_0 N(d_1) - K e^{-rT} N(d_2)$$

where

$$d_{1,2} := d_{+,-} := \frac{\log(S_0 e^{rT}/K)}{\sigma\sqrt{T}} \pm \frac{\sigma\sqrt{T}}{2}$$

ATM option prices are almost linear in vol



ATMF option price is

$$S_0 N(d_1) - Ke^{-rT} N(d_2) = S_0 (N(\sigma\sqrt{T}/2) - N(-\sigma\sqrt{T}/2))$$

For small $|x|$, $x_0 = 0$

$$N(x) = N(0) + N'(0)x + \frac{1}{2}N''(0)x^2 + O(x^3) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}}x + 0 + O(x^3).$$

$N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

So option price is approximately

$$S_0\left(\frac{1}{2} + \frac{\sigma\sqrt{T}/2}{\sqrt{2\pi}} - \frac{1}{2} + \frac{\sigma\sqrt{T}/2}{\sqrt{2\pi}}\right) = \frac{S_0\sigma\sqrt{T}}{\sqrt{2\pi}} \approx 0.4 \times S_0\sigma\sqrt{T}.$$

Your answer: 8 dollars

(True answer: 7.97 dollars)

Follow-up question: ATM delta

Same assumptions. What's the delta of the ATMF call?

Differentiating the approximate option price $S_0\sigma\sqrt{T}/\sqrt{2\pi}$ with respect to S_0, we have a delta of

$$\frac{\sigma\sqrt{T}}{\sqrt{2\pi}} \approx 0.08$$

Or do we? I thought the delta of an ATMF call should be close to 0.5.

$r=0$

$C(S,K)$

$$\frac{d}{dS}C(S,S)\Big|_{S=S_0}$$

$$\boxed{\frac{\partial}{\partial S}C(S,K)\Big|_{S=K=S_0}} \rightarrow \text{This is ATM delta.}$$

Normal approximation

Pretend that S_T is normally distributed (rather than lognormally).

In particular, pretend that the *simple* return on the forward

$$\frac{F_T}{F_0} - 1 = \frac{S_T}{F_0} - 1 \quad \sim \quad \text{Normal}(0, \sigma\sqrt{T})$$

ATMF option on no-div stock has price

$$\begin{aligned} e^{-rT}\mathbb{E}(S_T - F_0)^+ &= \frac{e^{-rT}}{2}\mathbb{E}|S_T - F_0| = \frac{F_0 e^{-rT}}{2}\mathbb{E}\left|\frac{S_T}{F_0} - 1\right| = \frac{S_0}{2}\mathbb{E}\left|\frac{S_T}{F_0} - 1\right| \\ &= \frac{1}{\sqrt{2\pi}} S_0 \sigma \sqrt{T} \end{aligned}$$

because the mean absolute deviation of a normal random variable is $\sqrt{\frac{2}{\pi}} \times$ its standard deviation (by calculating $\int_{-\infty}^{\infty} \frac{|x|}{\sqrt{2\pi}} e^{-x^2/2} \mathrm{d}x$)